Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

Benzene (Benzene)
Basis set   : external-CI
Charge   : 0
Multiplicity : 1
Total e-   : 6
n-body order : 2

GROUND STATE ENERGIES

E0 (Ha) : nan
E1 (Ha) : nan
Gap (Ha) : nan
HF (Ha) : nan

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 20

C_LB : coherence witness bound
C_gcg: Conjugate Gradient(Alg. A)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,6)

Frozen core orbs : 0
Frozen e-   : 0
Active e-   : 6
Active orbs   : 6
Spin-orbitals   : 12
Hilbert dim   : 400

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 1.058749
<D> / site : 0.176458

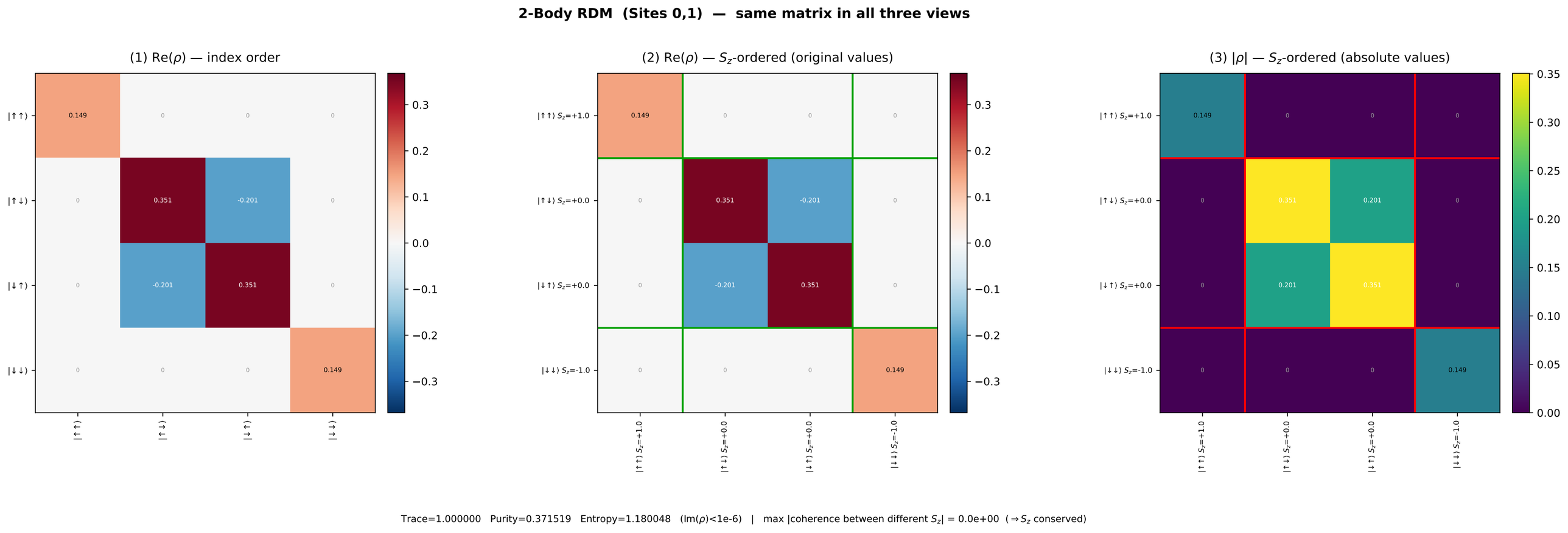
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Body order : 2
Clusters   : 15

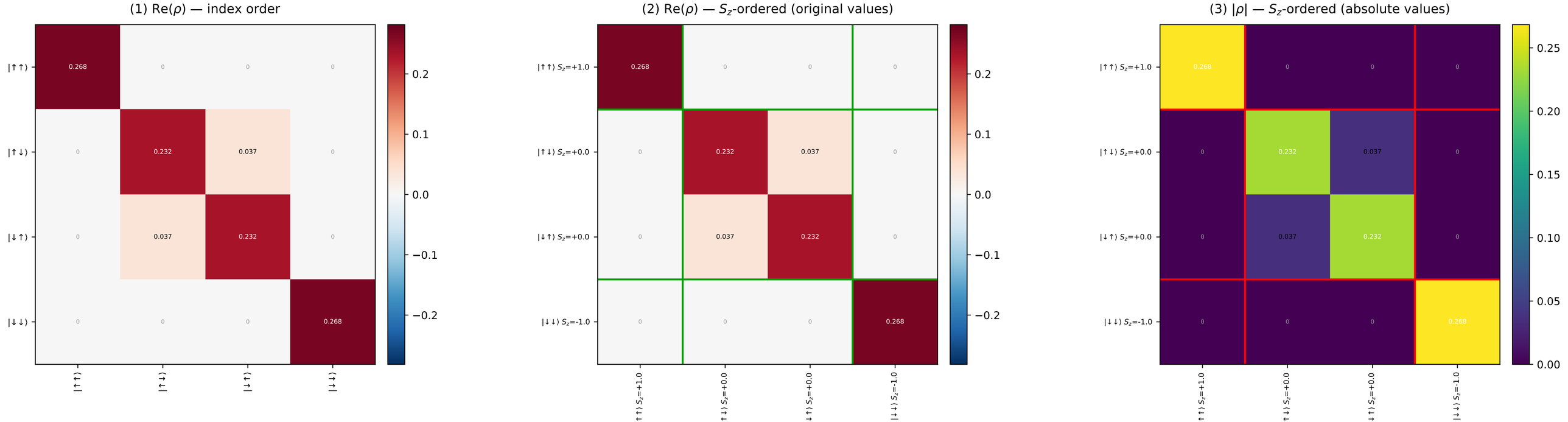
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Summary — 2-Body RDMs (Benzene, CAS(6,6))

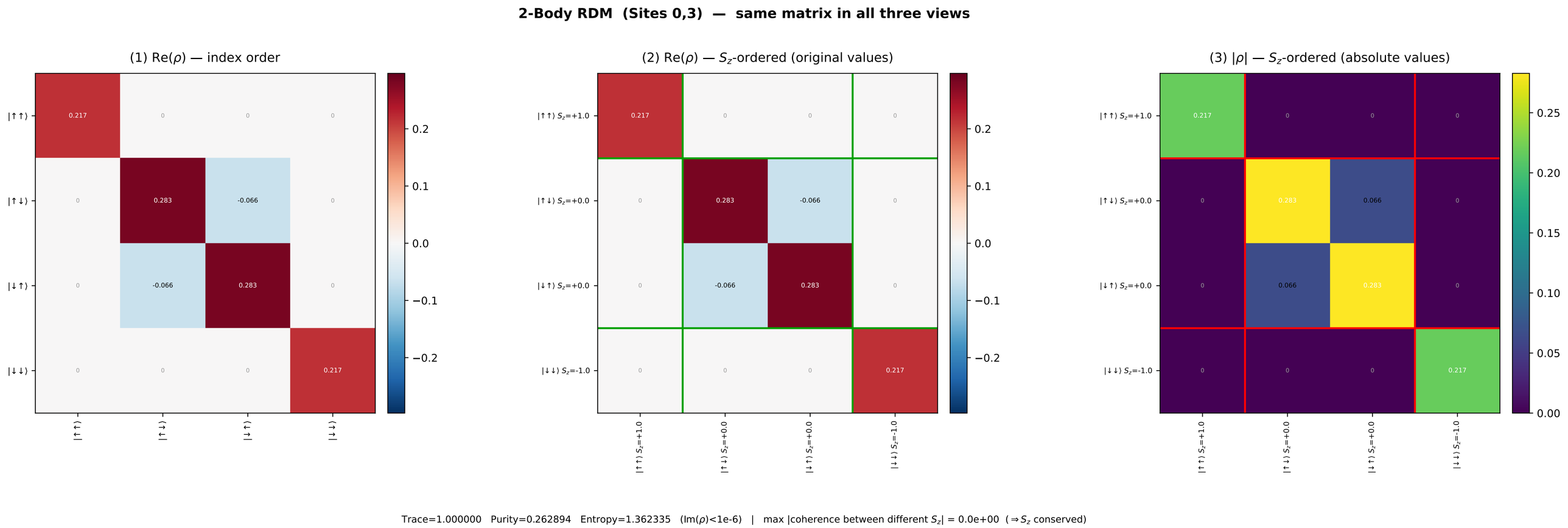
Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1	0.371519	1.180048	—	—	—	—	MIXED
0,2	0.254034	1.377774	—	—	—	—	MIXED
0,3	0.262894	1.362335	—	—	—	—	MIXED
0,4	0.254034	1.377774	—	—	—	—	MIXED
0,5	0.371503	1.180074	—	—	—	—	MIXED
1,2	0.371512	1.180059	—	—	—	—	MIXED
1,3	0.254034	1.377774	—	—	—	—	MIXED
1,4	0.262894	1.362335	—	—	—	—	MIXED
1,5	0.254034	1.377774	—	—	—	—	MIXED
2,3	0.371516	1.180053	—	—	—	—	MIXED
2,4	0.254034	1.377774	—	—	—	—	MIXED
2,5	0.262894	1.362335	—	—	—	—	MIXED
3,4	0.371512	1.180059	—	—	—	—	MIXED
3,5	0.254034	1.377774	—	—	—	—	MIXED
4,5	0.371519	1.180048	—	—	—	—	MIXED


2-Body RDM (Sites 0,1) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑⟩ S_z=+1.0
|↑↓⟩ S_z=+0.0
|↓↑⟩ S_z=+0.0
|↓↓⟩ S_z=-1.0
0.149
0.351
-0.201
0.201
Trace=1.000000 Purity=0.371519 Entropy=1.180048 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒S_z conserved)

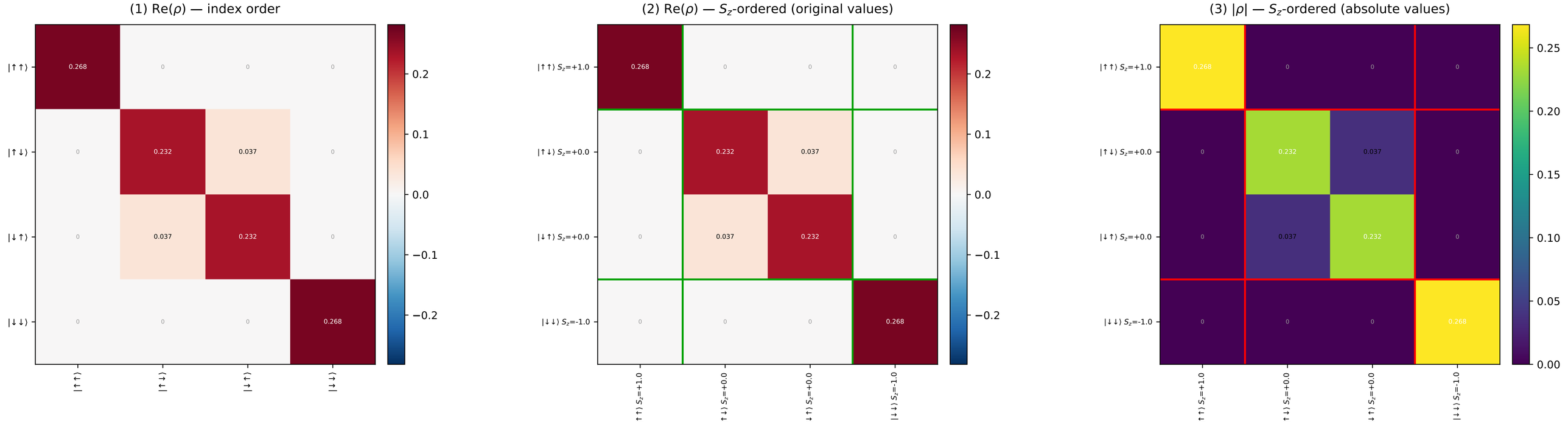
2-Body RDM (Sites 0,2) — same matrix in all three views



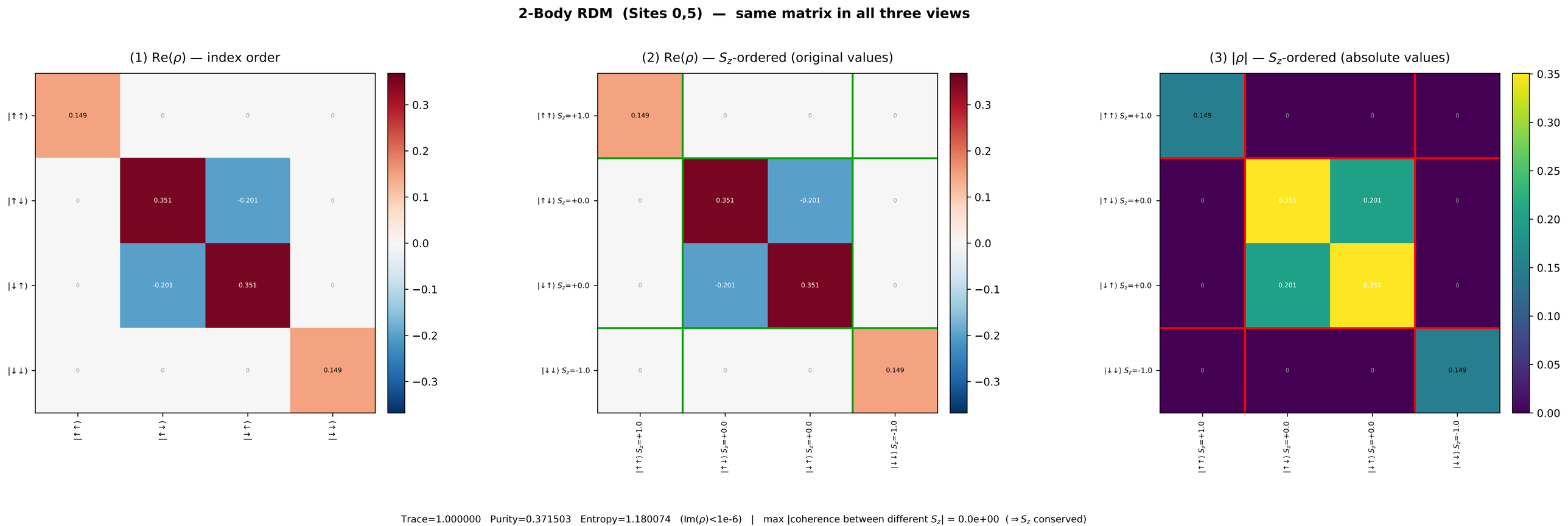
Trace=1.000000 Purity=0.254034 Entropy=1.377774 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


2-Body RDM (Sites 0,3) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑⟩ S_z=+1.0
|↑↓⟩ S_z=+0.0
|↓↑⟩ S_z=+0.0
|↓↓⟩ S_z=-1.0
0.217
0.283
-0.066
0.066
Trace=1.000000 Purity=0.262894 Entropy=1.362335 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒S_z conserved)

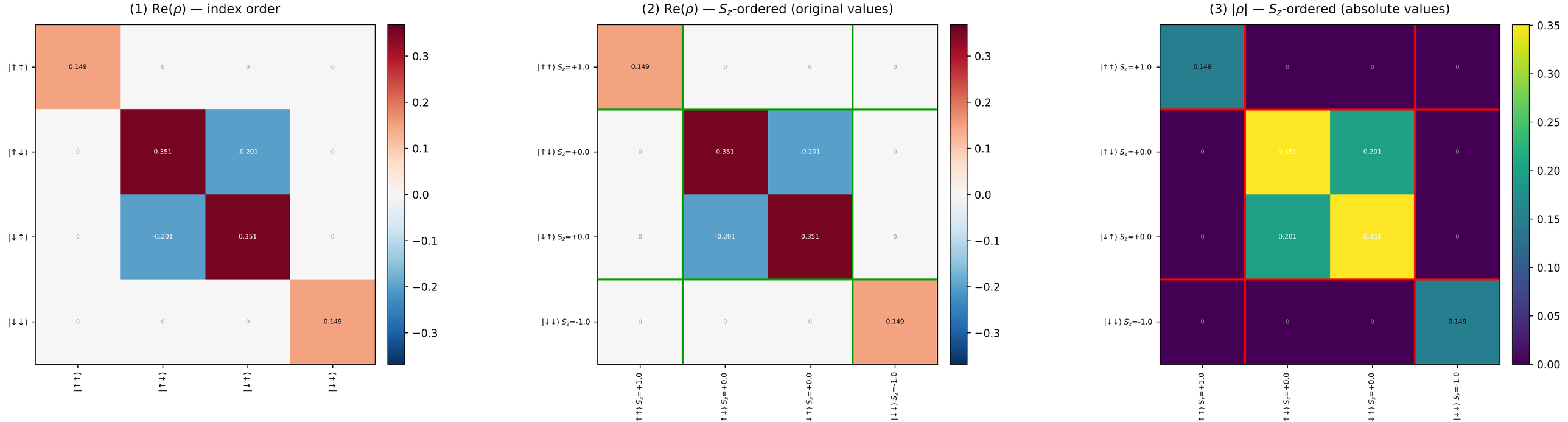
2-Body RDM (Sites 0,4) — same matrix in all three views



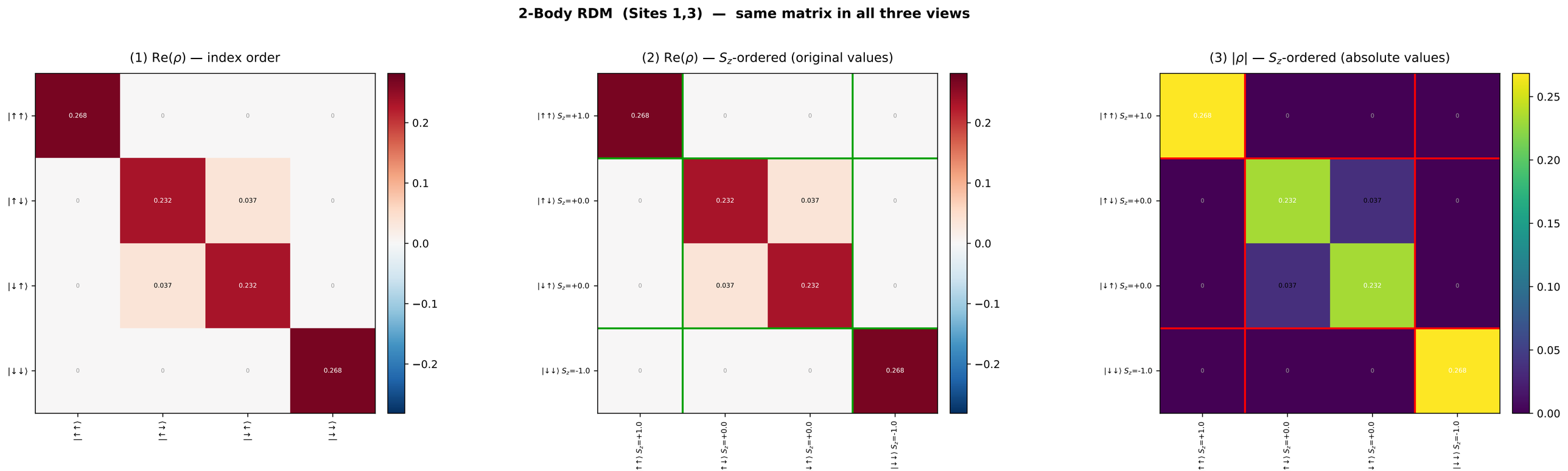
Trace=1.000000 Purity=0.254034 Entropy=1.377774 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


2-Body RDM (Sites 0,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑⟩ S_z=+1.0
|↑↓⟩ S_z=+0.0
|↓↑⟩ S_z=+0.0
|↓↓⟩ S_z=-1.0
0.149
0.351
-0.201
0.201
Trace=1.000000 Purity=0.371503 Entropy=1.180074 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒S_z conserved)

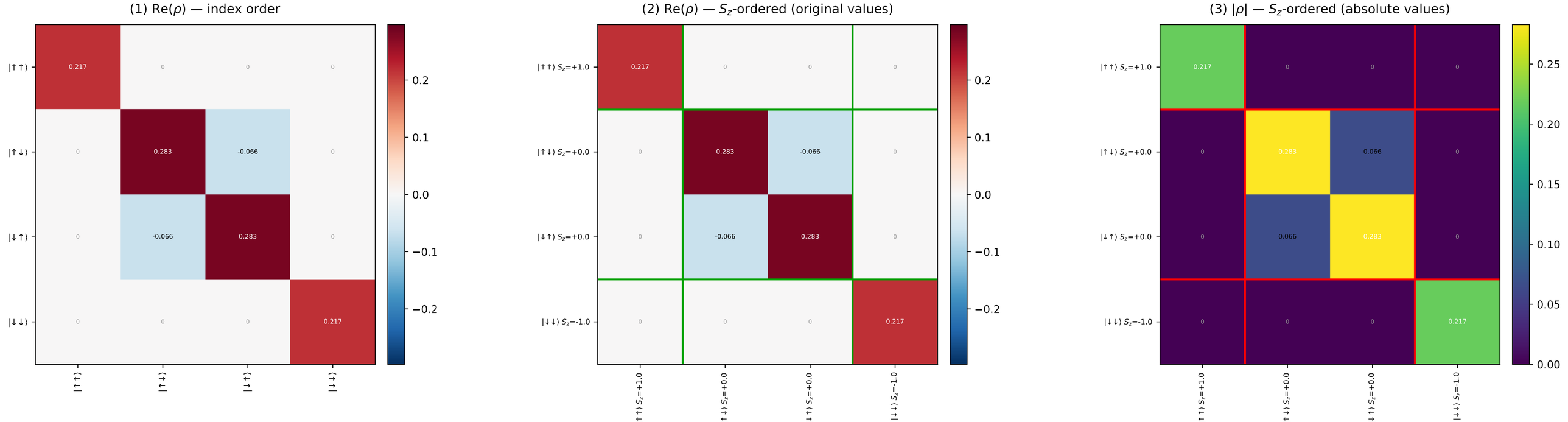
2-Body RDM (Sites 1,2) — same matrix in all three views



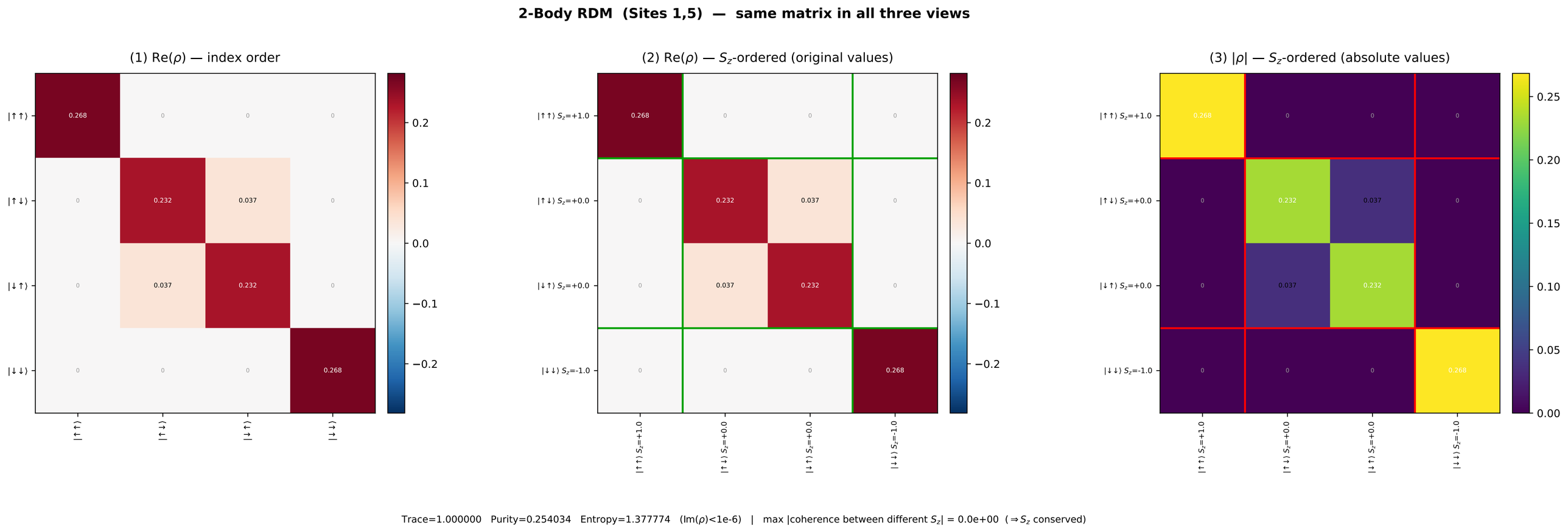
Trace=1.000000 Purity=0.371512 Entropy=1.180059 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


2-Body RDM (Sites 1,3) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑⟩ S_z=+1.0
|↑↓⟩ S_z=+0.0
|↓↑⟩ S_z=+0.0
|↓↓⟩ S_z=-1.0
0.268
0.232
0.037
Trace=1.000000 Purity=0.254034 Entropy=1.377774 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

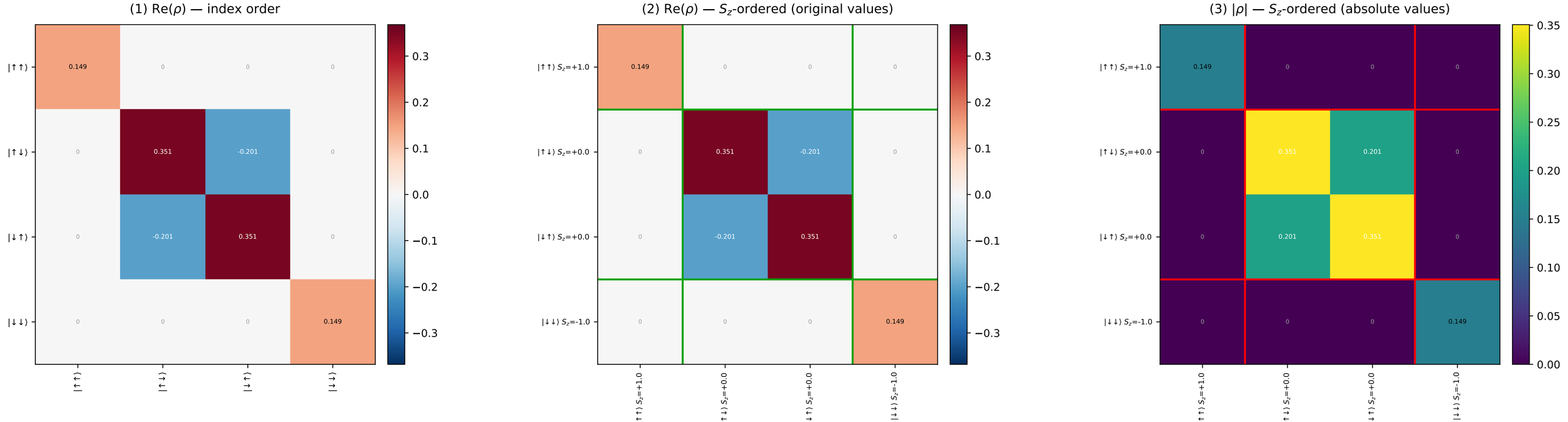
2-Body RDM (Sites 1,4) — same matrix in all three views



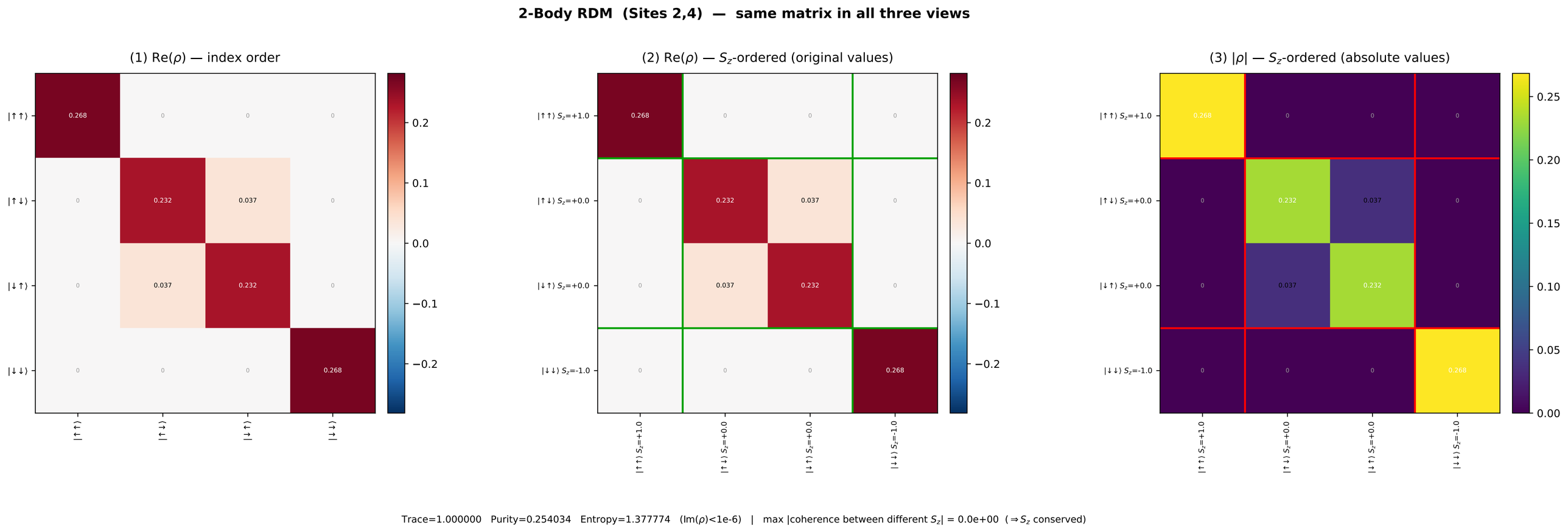
Trace=1.000000 Purity=0.262894 Entropy=1.362335 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

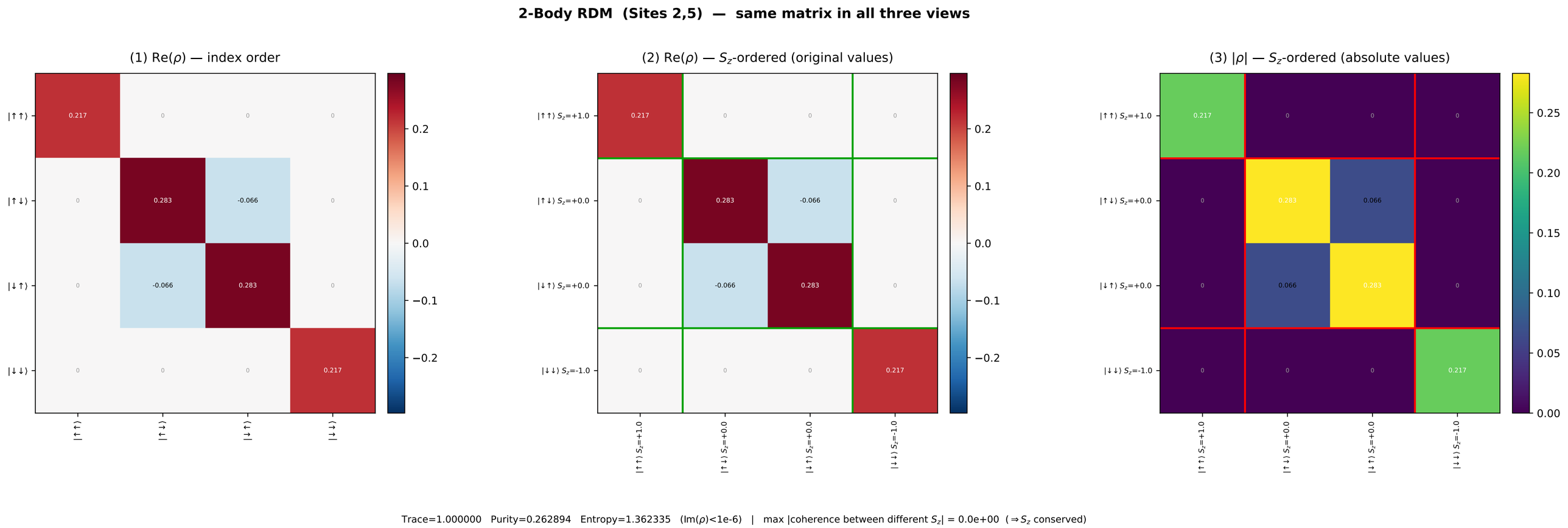

2-Body RDM (Sites 1,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑⟩
|↑↓⟩
|↓↑⟩
|↓↓⟩
S_z=+1.0
S_z=+0.0
S_z=+0.0
S_z=-1.0
0.268
0.232
0.037
0
0.2
0.1
0.0
−0.1
−0.2
0.25
0.20
0.15
0.10
0.05
0.00
Trace=1.000000 Purity=0.254034 Entropy=1.377774 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

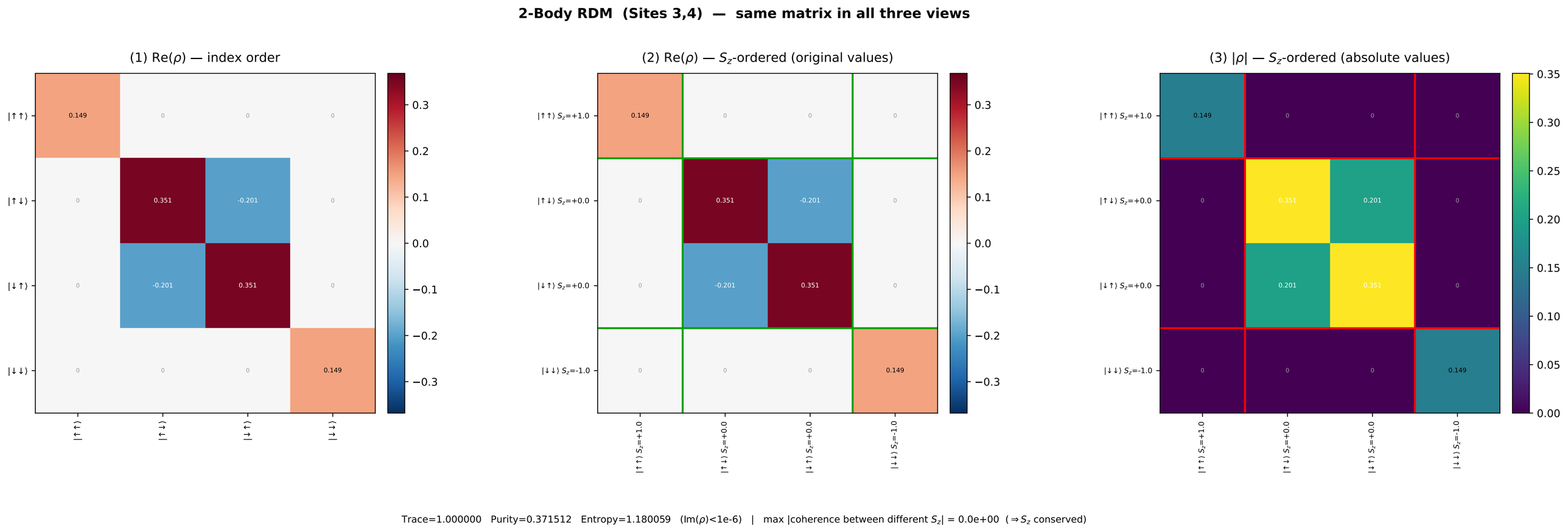
2-Body RDM (Sites 2,3) — same matrix in all three views

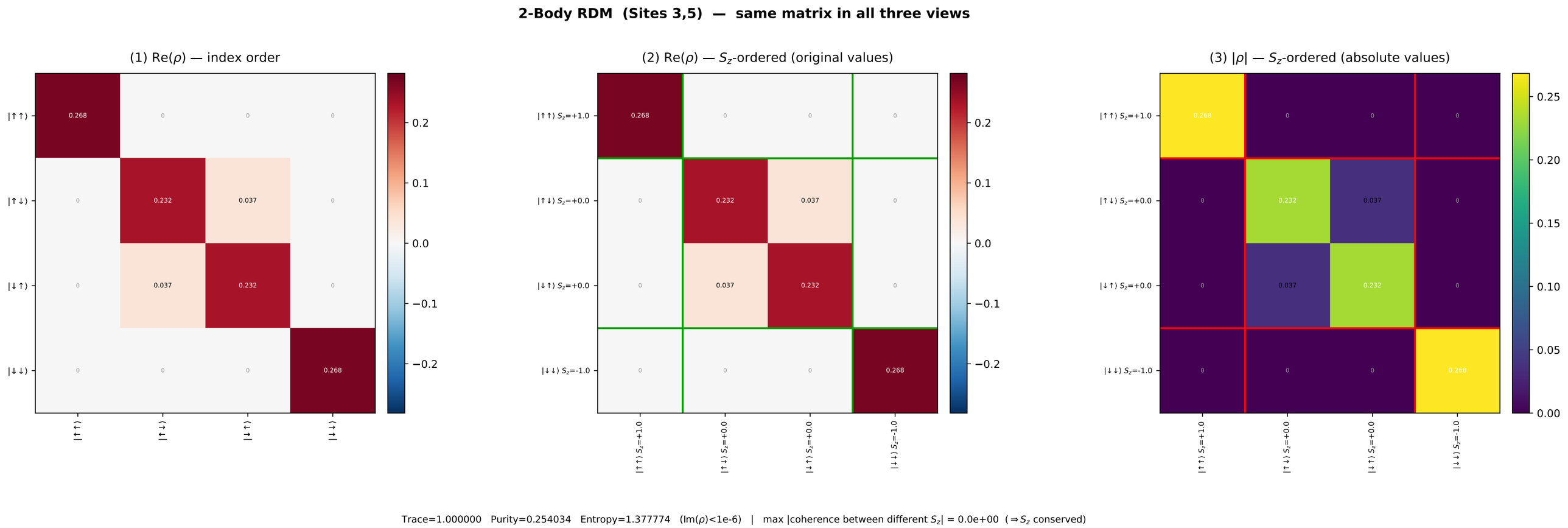


Trace=1.000000 Purity=0.371516 Entropy=1.180053 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

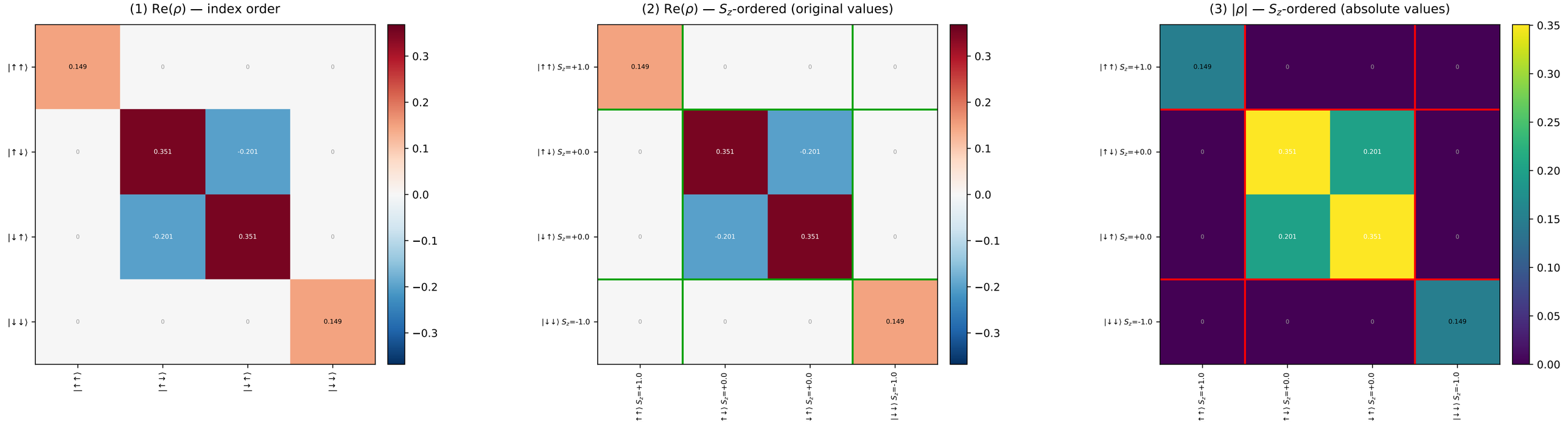

2-Body RDM (Sites 2,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑⟩ S_z=+1.0
|↑↓⟩ S_z=+0.0
|↓↑⟩ S_z=+0.0
|↓↓⟩ S_z=-1.0
0.268
0.232
0.037
0
Trace=1.000000 Purity=0.254034 Entropy=1.377774 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)


2-Body RDM (Sites 2,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑⟩ S_z=+1.0
|↑↓⟩ S_z=+0.0
|↓↑⟩ S_z=+0.0
|↓↓⟩ S_z=-1.0
0.217
0.283
-0.066
0.066
Trace=1.000000 Purity=0.262894 Entropy=1.362335 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒S_z conserved)


2-Body RDM (Sites 3,4) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑⟩ S_z=+1.0
|↑↓⟩ S_z=+0.0
|↓↑⟩ S_z=+0.0
|↓↓⟩ S_z=-1.0
0.149
0.351
-0.201
0.201
Trace=1.000000 Purity=0.371512 Entropy=1.180059 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒S_z conserved)


2-Body RDM (Sites 3,5) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
|↑↑⟩ S_z=+1.0
|↑↓⟩ S_z=+0.0
|↓↑⟩ S_z=+0.0
|↓↓⟩ S_z=-1.0
0.268
0.232
0.037
0
Trace=1.000000 Purity=0.254034 Entropy=1.377774 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

2-Body RDM (Sites 4,5) — same matrix in all three views



Trace=1.000000 Purity=0.371519 Entropy=1.180048 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)